

Efficient Framework for Hidden Camera Detection & Jamming using IoT

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Abstract— Hidden Camera so called “Spy Camera” is frequently used to record something without prior knowledge. Usually, the hidden camera will not be visible for objects that are recorded. The use of hidden cameras in public spaces has rapidly increased, and correspondingly the hidden camera recordings may be shared over a network. This research study has proposed a IoT-based architecture for performing camera detection and jamming. The client recognizes the network, where the camera is recording/streaming and jams the camera linked to that network through IoT. The cameras connected to the entire network may be jammed here. An extension to this RFID is introduced to identify the hidden camera signals and forecast the presence of a camera in that location.

Keywords— Node MCU, Microcontroller, Detector, Jamming.

I. INTRODUCTION

There will be chance of hidden camera installed in any place which will be used to capture our each and every action without our permission. The video that are recorded by the hidden camera will be used by some other people and they will misuse it. It will sometimes create a severe threat for human lives. The presence of hidden camera will create many

problems and it will affect the privacy of an individual. There must be a method to detect the hidden camera. The existence of a camera can be detected manually but it will be a more difficult operation. In other cases, detecting the camera is difficult if the buyers are unaware of the technological features. Some approaches for automatically detecting the camera will deliver a better output while also improving the detection ratio. Many detection approaches that have been presented for the identification of hidden cameras will deliver higher performance. The performance ratio in detection of the camera and stop the working of camera need to be improved. To stop using camera for bad, this research study has presented an idea of jamming the camera in restricted areas. In this if a camera is hidden and cannot be seen anywhere, this research study utilizes RFID to detect the signals and then the camera which is connected to a network can be jammed by using the IoT technology.

II. RELATED WORKS

The proposed system detects the presence of camera is based on the experimental survey on both the flicker and blink will detect the camera accurately [1]. The performed experimental work indicates that the hidden camera present in the room can be identified with a high accuracy rate [2]. More detailed information is given for providing a security to the household IoT devices[3]. Provided clear details about the existing threats and also given set of recommendations for improving the security features[4]. A high accuracy rate is achieved in audio transmission detection, which will provide a better effective method [5]. Protection will be provided to safeguard the infrastructure against certain drone attacks [6]. Survey is conducted on various papers focusing on cyber security[7]. IoT devices are used to find the water level in various water bodies in order to alert everyone in case of flood[8]. Efficient algorithms are used for finding the traffic jam as well as the accident that occurs in the particular location[9]. Security and privacy measures in IoT is discussed and provided a clear solution to overcome each and every attacks[10]. Framework proposed will give a better accuracy in terms of object detection and recognition of face[11]. Developed hardware for providing security in IoT and achieved a higher accuracy rate[12]. The work focus on managing the traffic in a critical situation with the use of IoT technique[13]. The microphones which is used for spying purpose can be detected and jammed[14]. Increased the detection Accuracy rate by using machine learning algorithms[15]. The major set of drawback of the existing system is that there is certain difficulty in the process. In some set of techniques, only the detection of camera is done so that we need to manually deactivate the camera. The other set of techniques for deactivation techniques are proposed with limited set of accuracy.

III. PROPOSED WORK

The Given Proposed System Is Done By Using Arduino Lilly pad microcontroller. Figure 1 explains the block diagram, the esp8266 is a Wi-Fi module to provide internet connection to an controller. The RF sensor is used to find the hidden camera. It is based on the frequency that is produced by the cameras that are present in that particular area. When an hidden camera is detected RF sensor is getting trigger and send the value to the cloud. By using cloud it send the alert message to the person and also to the family person and also gives the siren sound through the buzzer.

A. Hardware Used

- Arduino Lillypad
- Rf Detector
- Esp8266 Wifi Module

A radio frequency (RF) finder is a gadget used to distinguish the presence of RF waves either in a remote or wired actual transmission medium. They are otherwise called RF power indicators or RF reacting identifiers and are accessible as gadgets or modules. In the present innovative world, reconnaissance cameras are all over. Secret camera indicator applications are accessible on your Android or iPhone

or at a local security store, a secret camera finder can discover a camera anyplace regardless of how well it's covered up.

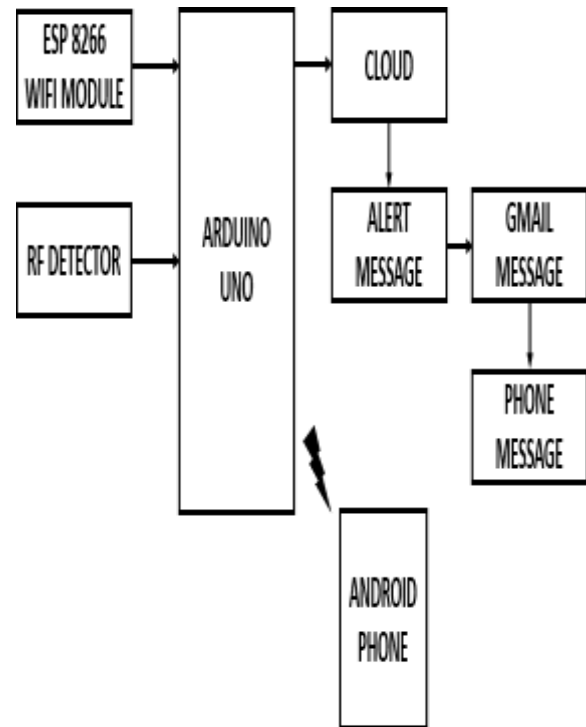


Fig. 1. Block Diagram

IV. RESULTS

Figure 2 describes the interconnection of hardware devices to generate the output. The devices such as Arduino, RF detector and other connecting cables are shown in the figure. Figure 3 describes the client side information and Figure 4 explains the server side, where all the details are given. Figure 5 indicates the detection of camera once the camera is detected to give the indication. Based on the input data received, it will analyse and check whether any camera is available in that particular room. If the system finds any camera it will give the alert through the hardware. After receiving the alert from the hardware the system needs to jam the camera. Figure 6 visualizes the jamming of camera once the camera is detected. The power that will be utilized by the framework will be less so that it will be a better option to go with this set of technique. Compared to the existing system, the performance gets improved. It is a better way for finding the undetected camera that is fixed inside the room. It will not only consider detection but also provide the mechanism for jamming the camera.

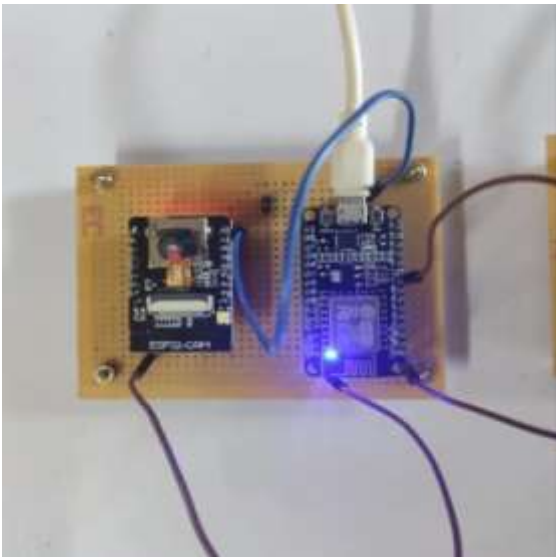


Fig. 2. Interconnection of hardware devices

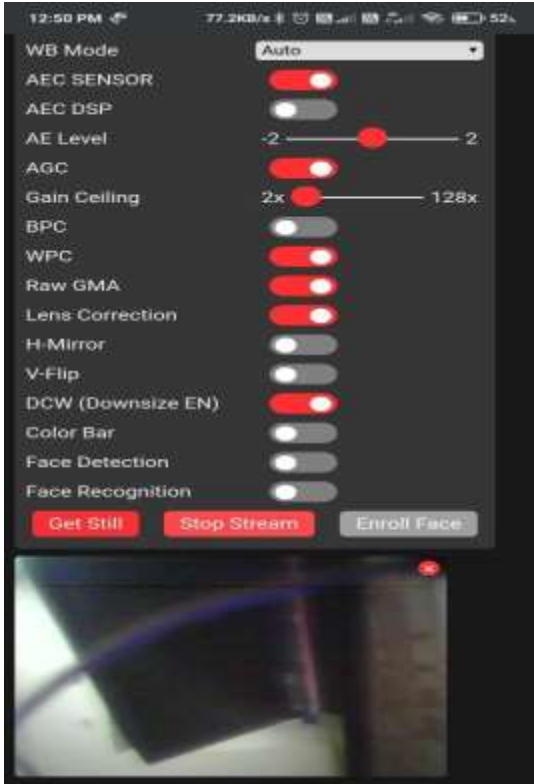


Fig. 4. Server Side

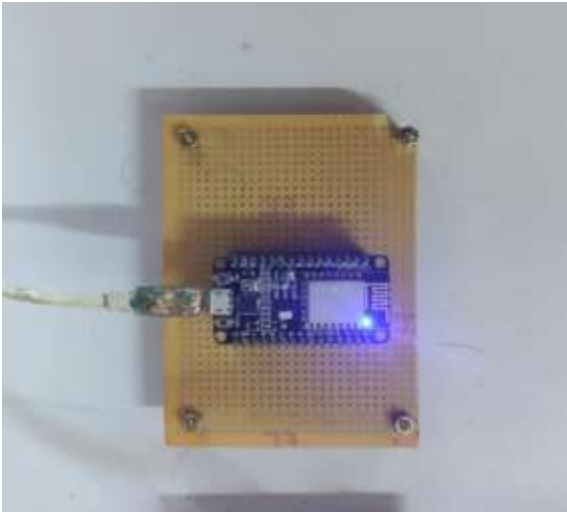


Fig. 3. Client Side

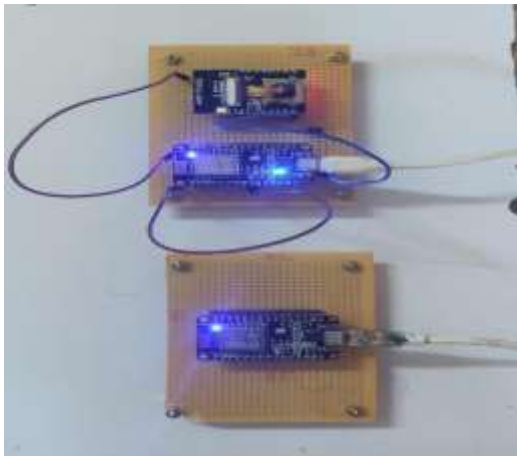


Fig. 5. Detection of Camera

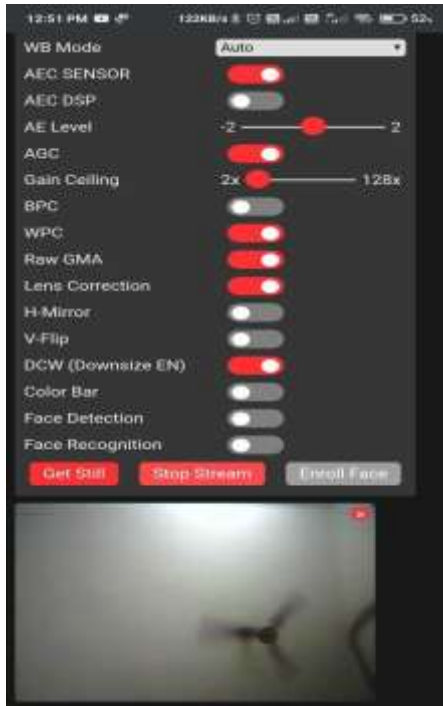


Fig. 6. Jamming the Camera

V. CONCLUSION

This paper describes a framework for the detection and jamming the camera using IoT devices. The spy camera is a major threat to the privacy and that problem has been overcome with the help of the proposed framework. The main purpose of the proposed framework is to detect the spy camera installed in the room and to jam it. The proposed concept works better when compared to the existing methods and provides higher degree of accuracy.

REFERENCES

- [1] Liu, T., Liu, Z., Huang, J., Tan, R., & Tan, Z. (2018, June). Detecting wireless spy cameras via stimulating and probing. In *Proceedings of the 16th Annual International Conference on Mobile Systems, Applications, and Services* (pp. 243-255).
- [2] Cheng, Y., Ji, X., Lu, T., & Xu, W. (2019). On detecting hidden wireless cameras: A traffic pattern-based approach. *IEEE Transactions on Mobile Computing*, 19(4), 907-921.
- [3] OConnor, T. J., Enck, W., & Reaves, B. (2019, May). Blinded and confused: uncovering systemic flaws in device telemetry for smart-home internet of things. In *Proceedings*

- of the 12th Conference on Security and Privacy in Wireless and Mobile Networks (pp. 140-150).
- [4] Costin, A. (2016, October). Security of cctv and video surveillance systems: Threats, vulnerabilities, attacks, and mitigations. In *Proceedings of the 6th international workshop on trustworthy embedded devices* (pp. 45-54).
- [5] Mitev, R., Pazii, A., Miettinen, M., Enck, W., & Sadeghi, A. R. (2020, December). Leakypick: Iot audio spy detector. In *Annual Computer Security Applications Conference* (pp. 694-705).
- [6] Multerer, T., Ganis, A., Prechtel, U., Miralles, E., Meusling, A., Mietzner, J., ... & Ziegler, V. (2017, October). Low-cost jamming system against small drones using a 3D MIMO radar based tracking. In *2017 European Radar Conference (EURAD)* (pp. 299-302). IEEE.
- [7] Krishna, C. L., & Murphy, R. R. (2017, October). A review on cybersecurity vulnerabilities for unmanned aerial vehicles. In *2017 IEEE International Symposium on Safety, Security and Rescue Robotics (SSRR)* (pp. 194-199). IEEE.
- [8] Pravin, A., Jacob, T. P., & Rajakumar, R. (2021, July). Enhanced Flood Detection System using IoT. In *2021 6th International Conference on Communication and Electronics Systems (ICCES)* (pp. 507-510). IEEE.
- [9] Kamijo, S., Matsushita, Y., Ikeuchi, K., & Sakauchi, M. (2000). Traffic monitoring and accident detection at intersections. *IEEE transactions on Intelligent transportation systems*, 1(2), 108-118.
- [10] Dabbagh, M., & Rayes, A. (2019). Internet of things security and privacy. In *Internet of Things from hype to reality* (pp. 211-238). Springer, Cham.
- [11] Jacob, T. P., Pravin, A., & Rajakumar, R. (2021, July). An AI-Powered Smart Camera for Object Detection. In *2021 6th International Conference on Communication and Electronics Systems (ICCES)* (pp. 1701-1703). IEEE.
- [12] Stankovic, J. A., Le, T., Hendawi, A., & Tian, Y. (2019). Hardware/software security patches for internet of trillions of things. *arXiv preprint arXiv:1903.05266*.
- [13] Rajesh, J., Naveen, I., Pravin, A., Jacob, T. P., & Nagarajan, G. (2020, March). Intelligent Traffic Management in Emergency Situations. In *International Conference on Emerging Trends and Advances in Electrical Engineering and Renewable Energy* (pp. 499-507). Springer, Singapore.
- [14] Roy, N., Hassanieh, H., & Roy Choudhury, R. (2017, June). Backdoor: Making microphones hear inaudible sounds. In *Proceedings of the 15th Annual International Conference on Mobile Systems, Applications, and Services* (pp. 2-14).
- [15] De Miguel, K., Brunete, A., Hernando, M., & Gambao, E. (2017). Home camera-based fall detection system for the elderly. *Sensors*, 17(12), 2864.